Conformal Risk Prediction

Distribution-Free Uncertainty Quantification for Financial Risk

Andrés González Ortega

March 2026

Abstract

An applied treatment of conformal prediction methods for uncertainty quantification in financial risk models. Point estimates from machine-learning models — a probability of default of 0.12, a Value-at-Risk of $2,000 — are routinely used for consequential decisions in credit underwriting and capital allocation, yet they are presented without any finite-sample guarantee on their accuracy. Traditional uncertainty quantification relies on the delta method (requires Gaussianity), the bootstrap (requires i.i.d. sampling), or parametric intervals (requires distributional assumptions) — all of which fail when their premises break down. Conformal prediction provides *distribution-free, finite-sample* coverage guarantees under a single, mild assumption: exchangeability of the data. We develop the framework from first principles: split conformal prediction for regression; Conformalized Quantile Regression (CQR) for heteroscedastic intervals; Conformal Risk Control for general monotone risk functionals; the main application to conformal PD intervals for credit scoring models (bridging Project 03); and Adaptive Conformal Inference for non-stationary financial time series. A worked example runs throughout: an XGBoost credit model whose PD point estimates are wrapped in prediction intervals with rigorous $1 - \alpha$ coverage guarantees. We assume familiarity with supervised learning, quantile regression, and basic probability theory.

Contents

1 The Uncertainty Problem in Risk **3**
1.1 Why Point Estimates Are Dangerous . . . 3
1.2 Traditional Approaches and Their Limitations . . . 3
1.3 The Conformal Promise . . . 4

2 Split Conformal Prediction **4**
2.1 Setup . . . 4
2.2 The Algorithm . . . 5
2.3 Why It Works . . . 5
2.4 Worked Example: Conformal Intervals for Daily Returns . . . 5

3 Conformalized Quantile Regression (CQR) **6**
3.1 Quantile Regression Recap . . . 6
3.2 The CQR Algorithm . . . 6
3.3 Why the Score Works . . . 7
3.4 Application: VaR Prediction Intervals . . . 7

4 Conformal Risk Control (CRC) **8**
4.1 Setup . . . 8
4.2 The CRC Procedure . . . 9
4.3 Application: Controlling Expected Shortfall . . . 9

5 ML Model Uncertainty: Conformal PD Intervals **9**
5.1 The Problem . . . 10
5.2 Method: Split Conformal for Classification . . . 10
5.3 Worked Example . . . 10
5.4 Mondrian Conformal Prediction for PD . . . 11
5.5 Practical Considerations . . . 12

6 Adaptive Conformal Inference (ACI) **12**
6.1 The Problem: Non-Stationarity . . . 12
6.2 The ACI Algorithm . . . 12
6.3 Intuition Behind the Update Rule . . . 13
6.4 Theoretical Guarantee . . . 13
6.5 Demonstration: Volatility Regime Shift . . . 14
6.6 Practical Choices . . . 14

7 References **14**

1 The Uncertainty Problem in Risk

A risk model produces a number. A Value-at-Risk model says "your 1-day 99% VaR is \$2,000." An ML credit scoring model outputs "the probability of default for this applicant is PD = 0.12." A portfolio optimizer recommends "allocate 40% to equities."

These are *point estimates*. They carry no statement about how wrong they might be.

1.1 Why Point Estimates Are Dangerous

Decisions downstream of risk models are binary and consequential:

- Approve or reject a loan application.
- Allocate \$X million in regulatory capital.
- Trigger a margin call or not.

A PD of 0.12 might lead to approval; a PD of 0.18 might trigger rejection. If the model's uncertainty band spans $[0.08, 0.18]$, the decision is not clear-cut — and the decision-maker *deserves to know this*.

Warning

Model risk regulation (SR 11-7 in the US, SS1/23 in the UK) explicitly requires that institutions quantify and communicate model uncertainty. The EU AI Act further demands that high-risk AI systems — including credit scoring — provide "appropriate measures of uncertainty" to users. A point estimate without an uncertainty band violates the spirit of all three frameworks.

1.2 Traditional Approaches and Their Limitations

Three classical strategies attempt to put error bars on model predictions:

1. **Delta method.** Approximate the estimator's distribution as Gaussian using a Taylor expansion around the true parameter. The resulting confidence interval is $\hat{\theta} \pm z_{1-\alpha/2} \cdot \widehat{\mathrm{se}}(\hat{\theta})$. *Problem:* requires the estimator to be asymptotically normal and the sample size to be "large enough" (an unverifiable assumption). Fails for tree ensembles, neural networks, and other non-smooth models.

2. **Bootstrap.** Resample the training data B times, refit the model each time, and compute an empirical distribution of predictions. *Problem:* assumes the original sample is i.i.d. from the population. Financial data exhibit serial dependence, regime changes, and concept drift. The bootstrap can also be computationally prohibitive for large models (B refits of an XGBoost with 1,000 trees is expensive).

3. **Parametric intervals.** Assume the prediction error follows a known distribution (Gaussian, Student-t) and construct intervals from its quantiles. *Problem:* model residuals are often heavy-tailed, heteroscedastic, and non-Gaussian. Misspecifying the error distribution invalidates the coverage guarantee.

Core Idea

All three classical methods require **assumptions that cannot be verified from the data**: Gaussianity, i.i.d. sampling, or a parametric error distribution. When these assumptions fail — as they routinely do in finance — the stated coverage level (e.g., 95%) is a fiction. Conformal prediction replaces these unverifiable assumptions with a single, testable condition: **exchangeability**.

1.3 The Conformal Promise

Conformal prediction provides the following guarantee:

Theorem 1.1 (Conformal coverage guarantee, informal)**.** *Let* $(X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, Y_{n+1})$ *be exchangeable random variables. Let* $\mathcal{C}(X_{n+1})$ *be the prediction set constructed by the split conformal procedure at level* α*. Then:*

$$\mathbb{P}\big(Y_{n+1} \in \mathcal{C}(X_{n+1})\big) \;\geq\; 1 - \alpha. \tag{1}$$

This holds for ***any*** *sample size* n*,* ***any*** *model* f*, and* ***any*** *joint distribution of* (X, Y)*.*

No asymptotics. No distributional assumptions beyond exchangeability. The guarantee holds for $n = 50$ just as it holds for $n = 50{,}000$.

Definition 1.1 (Exchangeability)**.** Random variables $Z_1, \ldots, Z_m$ are **exchangeable** if for every permutation π of $\{1, \ldots, m\}$:

$$(Z_1, \ldots, Z_m) \;\stackrel{d}{=}\; (Z_{\pi(1)}, \ldots, Z_{\pi(m)}). \tag{2}$$

Exchangeability is strictly weaker than the i.i.d. assumption: i.i.d. implies exchangeability, but exchangeability allows for dependence (e.g., draws from a Pólya urn are exchangeable but not independent).

2 Split Conformal Prediction

We now present the core algorithm in its simplest form.

2.1 Setup

We have a dataset $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^N$ where $X_i \in \mathcal{X} \subseteq \mathbb{R}^d$ are features and $Y_i \in \mathcal{Y} \subseteq \mathbb{R}$ is the response (e.g., next-day return, realized loss, default indicator).

We split $\mathcal{D}$ into two disjoint sets:

- **Training set** $\mathcal{D}_{\text{train}}$ of size m: used to fit the predictive model $\hat{f}$.
- **Calibration set** $\mathcal{D}_{\text{cal}}$ of size $n = N - m$: used to compute nonconformity scores. The calibration set is *never* used during model training.

2.2 The Algorithm

Algorithm 1: Split Conformal Prediction

Input: Training data $\mathcal{D}_{\text{train}}$, calibration data $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$, miscoverage level $\alpha \in (0, 1)$, new input X_{n+1}.

1. Train model $\hat{f}$ on $\mathcal{D}_{\text{train}}$.
2. Compute nonconformity scores on $\mathcal{D}_{\text{cal}}$:
$$s_i = \left|Y_i - \hat{f}(X_i)\right|, \qquad i = 1, \ldots, n.$$
3. Compute the conformal quantile:
$$\hat{q} = \text{Quantile}\left(\{s_1, \ldots, s_n\};\ \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right).$$
4. Return the prediction interval:
$$\mathcal{C}(X_{n+1}) = \left[\hat{f}(X_{n+1}) - \hat{q},\ \hat{f}(X_{n+1}) + \hat{q}\right].$$

2.3 Why It Works

The key insight is that under exchangeability, the rank of the new score s_{n+1} among $\{s_1, \ldots, s_n, s_{n+1}\}$ is uniformly distributed over $\{1, \ldots, n+1\}$.

Theorem 2.1 (Finite-sample coverage)**.** *If $(X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}, Y_{n+1})$ are exchangeable and $\hat{f}$ is trained on a separate training set (independent of the calibration data), then:*

$$1 - \alpha \ \leq\ \mathbb{P}\big(Y_{n+1} \in \mathcal{C}(X_{n+1})\big) \ \leq\ 1 - \alpha + \frac{1}{n+1}. \tag{3}$$

Proof. The nonconformity scores $s_1, \ldots, s_n, s_{n+1}$ are exchangeable (since they are functions of exchangeable random variables applied through a fixed function $\hat{f}$, trained on a separate dataset). By exchangeability, the rank of s_{n+1} among all $n+1$ scores is uniformly distributed on $\{1, \ldots, n+1\}$. Therefore:

$$\mathbb{P}(s_{n+1} \leq \hat{q}) = \mathbb{P}(\text{rank}(s_{n+1}) \leq \lceil (n+1)(1-\alpha) \rceil) = \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1}.$$

Since $\lceil (n+1)(1-\alpha) \rceil \geq (n+1)(1-\alpha)$, we have $\mathbb{P}(s_{n+1} \leq \hat{q}) \geq 1 - \alpha$. The event $\{s_{n+1} \leq \hat{q}\}$ is exactly $\{Y_{n+1} \in \mathcal{C}(X_{n+1})\}$. The upper bound follows from $\lceil (n+1)(1-\alpha) \rceil \leq (n+1)(1-\alpha) + 1$. □

Key Insight

The coverage guarantee is **exact**: it holds for any finite n, any model $\hat{f}$, and any data distribution. The only price is a "rounding error" of at most $1/(n+1)$. With a calibration set of $n = 999$, the excess coverage is at most 0.1%.

2.4 Worked Example: Conformal Intervals for Daily Returns

Consider a linear model predicting next-day S&P 500 returns using 5 lagged returns as features. We split 2,000 observations: 1,200 for training, 800 for calibration. We target $\alpha = 0.05$ (95% coverage).

Example 2.1 (Return prediction intervals).

1. Train OLS on the training set: $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots + \hat{\beta}_5 X_5$.
2. Compute absolute residuals on the calibration set: $s_i = |Y_i - \hat{f}(X_i)|$ for $i = 1, \ldots, 800$.
3. Compute $\hat{q} = \text{Quantile}(\{s_1, \ldots, s_{800}\}; \lceil 801 \times 0.95 \rceil / 800) = \text{Quantile}(\{s_1, \ldots, s_{800}\}; 761/800)$. Suppose $\hat{q} = 1.42\%$.
4. For a new day with features X_{new}, suppose $\hat{f}(X_{\text{new}}) = 0.08\%$. The prediction interval is:

$$\mathcal{C}(X_{\text{new}}) = [0.08\% - 1.42\%,\ 0.08\% + 1.42\%] = [-1.34\%,\ 1.50\%].$$

On a held-out test set of 500 days, the conformal interval achieves **95.2% empirical coverage**, compared to 93.1% for the parametric normal-theory interval $\hat{f}(X) \pm 1.96 \cdot \hat{\sigma}$. The parametric interval under-covers because return residuals are heavy-tailed (excess kurtosis ≈ 4.8).

Warning

The conformal guarantee is **marginal**: $\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1 - \alpha$, averaged over the randomness of the calibration set and the new test point. It does *not* guarantee conditional coverage: $\mathbb{P}(Y \in \mathcal{C}(X) \mid X = x) \geq 1 - \alpha$ for every x. In regions where the model is poor, the interval may under-cover. In regions where the model is accurate, it may over-cover (be too wide). This motivates Conformalized Quantile Regression (Section 3).

3 Conformalized Quantile Regression (CQR)

Standard split conformal (Section 2) produces *symmetric* intervals of width $2\hat{q}$ centered at $\hat{f}(X)$. This is suboptimal when the conditional distribution of $Y \mid X$ is heteroscedastic or asymmetric — as is typical in finance.

VaR losses are right-skewed. Credit losses are zero-inflated. Equity returns have asymmetric tails. A symmetric interval wastes coverage budget on the wrong side.

3.1 Quantile Regression Recap

Quantile regression (Koenker & Bassett, 1978) directly estimates conditional quantiles. For a target quantile $\tau \in (0, 1)$, the quantile regression loss is:

$$\rho_\tau(u) = u \cdot \big(\tau - \mathbf{1}\{u < 0\}\big) = \begin{cases} \tau \cdot |u| & \text{if } u \geq 0, \\ (1 - \tau) \cdot |u| & \text{if } u < 0. \end{cases} \tag{4}$$

A model $\hat{q}_\tau(X)$ trained to minimize $\mathbb{E}[\rho_\tau(Y - \hat{q}_\tau(X))]$ estimates the τ-th conditional quantile of $Y \mid X$.

3.2 The CQR Algorithm

Romano, Patterson, and Candès (2019) combine quantile regression with conformal calibration to obtain adaptive, asymmetric prediction intervals with finite-sample coverage guarantees.

Algorithm 2: Conformalized Quantile Regression (CQR)

Input: Training data $\mathcal{D}_{\text{train}}$, calibration data $\mathcal{D}_{\text{cal}} = \{(X_i, Y_i)\}_{i=1}^n$, miscoverage level α, new input X_{n+1}.

1. Train two quantile models on $\mathcal{D}_{\text{train}}$:

$$\hat{q}_{\alpha/2}(X) \quad \text{(lower quantile)}, \qquad \hat{q}_{1-\alpha/2}(X) \quad \text{(upper quantile)}.$$

2. Compute conformity scores on $\mathcal{D}_{\text{cal}}$:

$$s_i = \max\big(\hat{q}_{\alpha/2}(X_i) - Y_i,\ Y_i - \hat{q}_{1-\alpha/2}(X_i)\big), \qquad i = 1, \ldots, n.$$

3. Compute the conformal adjustment:

$$\hat{Q} = \text{Quantile}\left(\{s_1, \ldots, s_n\};\ \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right).$$

4. Return the prediction interval:

$$\mathcal{C}_{\text{CQR}}(X_{n+1}) = \big[\hat{q}_{\alpha/2}(X_{n+1}) - \hat{Q},\ \hat{q}_{1-\alpha/2}(X_{n+1}) + \hat{Q}\big].$$

Core Idea

CQR uses quantile regression to capture the *shape* of the conditional distribution (asymmetry, heteroscedasticity) and then applies a conformal correction to guarantee finite-sample marginal coverage. The intervals are narrow where the model is confident and wide where it is uncertain — unlike standard conformal, which applies a single global width.

3.3 Why the Score Works

The CQR score $s_i = \max(\hat{q}_{\alpha/2}(X_i) - Y_i,\ Y_i - \hat{q}_{1-\alpha/2}(X_i))$ is the amount by which Y_i falls outside the estimated $[\alpha/2,\ 1-\alpha/2]$ interval. If Y_i lies inside the interval, the score is negative (the max of two negative numbers). The conformal quantile $\hat{Q}$ inflates the interval just enough so that at least $\lceil (n+1)(1-\alpha) \rceil$ calibration points fall inside.

Theorem 3.1 (CQR coverage)**.** *Under the same exchangeability assumption as Theorem 2.1, the CQR prediction interval satisfies:*

$$\mathbb{P}\big(Y_{n+1} \in \mathcal{C}_{\text{CQR}}(X_{n+1})\big) \ \geq\ 1 - \alpha. \tag{5}$$

The proof is identical to that of Theorem 2.1: the CQR scores are exchangeable, and the procedure selects the same quantile of these scores. The particular *form* of the score does not affect the coverage guarantee — only the adaptivity and efficiency of the resulting intervals.

3.4 Application: VaR Prediction Intervals

Traditional VaR is a single number: "the 99th percentile of the loss distribution is \$2,000." But VaR itself is estimated with uncertainty. CQR can provide a *prediction interval for VaR*:

Example 3.1 (VaR interval via CQR)**.** Let Y be the realized daily portfolio loss and X be the conditioning information (lagged returns, volatility estimates, market indicators).

1. Train gradient-boosted quantile regression models for $\tau = 0.005$ and $\tau = 0.995$ on the training set.

2. Compute CQR scores on the calibration set at level $\alpha = 0.05$.
3. For a new day, the quantile models produce $\hat{q}_{0.005}(X_{\text{new}}) = -\$3{,}100$ and $\hat{q}_{0.995}(X_{\text{new}}) = \$2{,}800$.
4. After conformal adjustment ($\hat{Q} = \$180$), the interval becomes $[-\$3{,}280,\ \$2{,}980]$.

Instead of reporting "$\text{VaR}_{99} = \$2{,}800$," we report "$\text{VaR}_{99} \in [\$2{,}620, \$2{,}980]$ with 95% conformal guarantee." This interval reflects the *estimation uncertainty* in VaR itself.

4 Conformal Risk Control (CRC)

Coverage — $\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1 - \alpha$ — is not the only risk measure that matters in finance. Regulators care about CVaR, expected shortfall, spectral risk measures, and tail-weighted loss functions. Conformal Risk Control (Angelopoulos et al., 2024) generalizes the conformal framework to control *any monotone risk functional.*

4.1 Setup

We have a family of prediction sets $\mathcal{C}_\lambda(X)$ indexed by a parameter $\lambda \in \Lambda \subseteq \mathbb{R}$. As λ increases, the prediction sets grow: $\lambda' \geq \lambda \implies \mathcal{C}_{\lambda'}(X) \supseteq \mathcal{C}_\lambda(X)$.

A **risk functional** $R : \Lambda \to [0, 1]$ measures the expected loss associated with prediction sets at threshold λ:

$$R(\lambda) = \mathbb{E}\big[\ell\big(Y, \mathcal{C}_\lambda(X)\big)\big], \tag{6}$$

where ℓ is a bounded loss function.

Definition 4.1 (Monotone risk)**.** The risk $R(\lambda)$ is **monotone non-increasing** in λ if larger prediction sets (higher λ) never increase the risk:

$$\lambda' \geq \lambda \implies R(\lambda') \leq R(\lambda). \tag{7}$$

Example 4.1 (Risk functionals in finance)**.**

Risk Functional	**Loss** ℓ	**Financial Use**
Miscoverage rate	$\mathbf{1}\{Y \notin \mathcal{C}_\lambda(X)\}$	Standard conformal (VaR)
Expected shortfall	$(Y - \sup \mathcal{C}_\lambda(X))^+$	CVaR / tail risk
False negative rate	$\mathbf{1}\{Y = 1,\ 1 \notin \mathcal{C}_\lambda(X)\}$	Missing a default
Interval width penalty	$\lvert\mathcal{C}_\lambda(X)\rvert \cdot \mathbf{1}\{Y \notin \mathcal{C}_\lambda(X)\}$	Efficiency-weighted coverage

4.2 The CRC Procedure

Algorithm 3: Conformal Risk Control

Input: Calibration data $\{(X_i, Y_i)\}_{i=1}^n$, risk level α, loss function ℓ, family $\{\mathcal{C}_\lambda\}_{\lambda \in \Lambda}$.

1. For each calibration point and each λ, compute:

$$\ell_i(\lambda) = \ell\big(Y_i, \mathcal{C}_\lambda(X_i)\big), \qquad i = 1, \ldots, n.$$

2. Compute the empirical risk:

$$\hat{R}_n(\lambda) = \frac{1}{n} \sum_{i=1}^n \ell_i(\lambda).$$

3. Select the threshold:

$$\hat{\lambda} = \inf\left\{ \lambda \in \Lambda : \hat{R}_n(\lambda) \le \alpha - \frac{B}{n+1} \right\},$$

   where $B = \sup_\lambda \ell(\cdot, \mathcal{C}_\lambda(\cdot))$ is the upper bound on the loss.

4. Return $\mathcal{C}_{\hat{\lambda}}(\cdot)$.

Theorem 4.1 (Conformal Risk Control guarantee (Angelopoulos et al., 2024))**.** *Let* $(X_1, Y_1), \ldots, (X_n, Y_n), (X_{n+1}$ *be exchangeable. If* $R(\lambda)$ *is monotone non-increasing in* λ *and* ℓ *is bounded in* $[0, B]$*, then:*

$$\mathbb{E}\big[\ell\big(Y_{n+1}, \mathcal{C}_{\hat{\lambda}}(X_{n+1})\big)\big] \;\le\; \alpha. \tag{8}$$

Key Insight

CRC subsumes standard conformal prediction as a special case (take ℓ to be the miscoverage indicator). But it also allows control of CVaR, expected shortfall, and any other monotone risk functional. The guarantee is finite-sample and distribution-free, just like standard conformal.

4.3 Application: Controlling Expected Shortfall

Suppose we want not just $\mathbb{P}(\text{loss} > \text{VaR}) \le \alpha$ (which is what VaR gives us) but also a bound on how bad losses can be *beyond* VaR. This is the domain of Expected Shortfall (ES):

$$\text{ES}_\alpha = \mathbb{E}\big[\text{Loss} \mid \text{Loss} > \text{VaR}_\alpha\big]. \tag{9}$$

CRC with loss $\ell(Y, \mathcal{C}_\lambda(X)) = (Y - \sup \mathcal{C}_\lambda(X))^+$ directly controls the expected tail overshoot. The threshold $\hat{\lambda}$ is calibrated so that the average exceedance beyond the prediction set boundary is at most α — a distribution-free ES guarantee.

5 ML Model Uncertainty: Conformal PD Intervals

This is the central application of the project, bridging the XGBoost credit scoring model from Project 03 with rigorous uncertainty quantification.

5.1 The Problem

A credit scoring model (logistic regression, random forest, XGBoost, neural network) outputs a probability of default $\widehat{\mathrm{PD}}(X)$ for an applicant with features X. This estimate is used to:

- Decide loan approval (reject if $\widehat{\mathrm{PD}} > \tau$).
- Price the loan (interest rate increases with PD).
- Compute regulatory capital (PD enters the Basel IRB formula).

But $\widehat{\mathrm{PD}}(X) = 0.12$ is a single number. Is the true PD closer to 0.08 or 0.18? The answer has material consequences: at 0.08 the loan is profitable; at 0.18 the expected loss exceeds the interest margin.

Read Before Coding

This section assumes you have read the Project 03 theory document on ML credit scoring and understand: (a) the binary classification setup ($Y \in \{0, 1\}$, default or not); (b) how XGBoost produces calibrated probability estimates; (c) the Brier score and other proper scoring rules.

5.2 Method: Split Conformal for Classification

For a binary classification problem, the nonconformity score measures how "surprised" the model is by the true outcome. We use:

$$s_i = \left|Y_i - \widehat{\mathrm{PD}}(X_i)\right|, \tag{10}$$

where $Y_i \in \{0, 1\}$ is the realized default indicator and $\widehat{\mathrm{PD}}(X_i)$ is the model's predicted probability.

- If $Y_i = 1$ (default) and $\widehat{\mathrm{PD}}(X_i) = 0.12$, then $s_i = |1 - 0.12| = 0.88$ (high score — the model is surprised).
- If $Y_i = 0$ (no default) and $\widehat{\mathrm{PD}}(X_i) = 0.12$, then $s_i = |0 - 0.12| = 0.12$ (low score — the model expected this).

The conformal prediction set for a new applicant is:

$$\mathcal{C}(X_{n+1}) = \left\{y \in [0, 1] : |y - \widehat{\mathrm{PD}}(X_{n+1})| \le \hat{q}\right\} = \left[\widehat{\mathrm{PD}}(X_{n+1}) - \hat{q},\ \widehat{\mathrm{PD}}(X_{n+1}) + \hat{q}\right] \cap [0, 1], \tag{11}$$

where $\hat{q}$ is the conformal quantile from the calibration set and we clip to $[0, 1]$ since PD is a probability.

5.3 Worked Example

Example 5.1 (Conformal PD intervals for XGBoost)**.** We have 10,000 loan applications with binary default labels ($Y \in \{0, 1\}$).

1. **Split:** 6,000 for training, 2,000 for calibration, 2,000 for testing.
2. **Train:** Fit an XGBoost classifier on the training set. The model outputs $\widehat{\mathrm{PD}}(X_i)$ for each applicant.
3. **Calibrate:** Compute scores on the calibration set: $s_i = |Y_i - \widehat{\mathrm{PD}}(X_i)|$ for $i = 1, \ldots, 2000$.

   The distribution of scores is bimodal:

   - Non-defaulters ($Y_i = 0$): scores cluster near $\widehat{\mathrm{PD}}(X_i)$, typically in $[0.02, 0.25]$.

- Defaulters ($Y_i = 1$): scores cluster near $1 - \widehat{\mathrm{PD}}(X_i)$, typically in $[0.75, 0.98]$.

4. **Threshold:** At $\alpha = 0.05$: $\hat{q} = \mathrm{Quantile}(\{s_1, \ldots, s_{2000}\}; 1901/2000) = 0.89$.

5. **Predict:** For a new applicant with $\widehat{\mathrm{PD}}(X_{\text{new}}) = 0.12$:

$$\mathcal{C}(X_{\text{new}}) = [0.12 - 0.89,\ 0.12 + 0.89] \cap [0, 1] = [0, 1].$$

Warning

The interval $[0, 1]$ is vacuous — it says nothing useful. This happens because the binary nature of $Y \in \{0, 1\}$ forces the scores for defaulters to be large ($\approx 1 - \widehat{\mathrm{PD}}$), inflating the 95th percentile. The fix is to use a **class-conditional** or **Mondrian** conformal approach, discussed next.

5.4 Mondrian Conformal Prediction for PD

The Mondrian approach (Vovk et al., 2005) computes separate conformal thresholds for subgroups. For credit scoring, the natural partition is by predicted risk band.

Definition 5.1 (Mondrian conformal)**.** Partition the calibration set into K groups $G_1, \ldots, G_K$ based on a grouping function $g(X)$. For each group k, compute a separate threshold:

$$\hat{q}_k = \mathrm{Quantile}\left(\{s_i : i \in G_k\};\ \frac{\lceil (|G_k| + 1)(1 - \alpha) \rceil}{|G_k|}\right). \tag{12}$$

For a new point with $g(X_{n+1}) = k$, use $\hat{q}_k$.

For credit scoring, we partition by PD decile and use a refined score. Rather than the absolute error against the binary label, we use the model's uncertainty directly:

Example 5.2 (Mondrian conformal PD intervals)**.** Partition calibration applicants into 5 groups by predicted PD: $[0, 0.05)$, $[0.05, 0.10)$, $[0.10, 0.20)$, $[0.20, 0.50)$, $[0.50, 1.0]$.

Within each group, compute the nonconformity score using a proper scoring rule residual. For the group $[0.10, 0.20)$ containing 400 calibration points with a default rate of 14%, the conformal threshold is $\hat{q}_3 = 0.063$.

For a new applicant with $\widehat{\mathrm{PD}} = 0.12$ (falling in group 3):

$$\mathcal{C}(X_{\text{new}}) = [0.12 - 0.063,\ 0.12 + 0.063] = [0.057,\ 0.183].$$

This is a **useful** interval: it tells the loan officer that the PD is somewhere between 5.7% and 18.3% with 95% confidence.

Key Insight

Conformal PD intervals satisfy two regulatory requirements simultaneously:

1. **SR 11-7 (Model Risk Management):** "Uncertainty about model outputs should be quantified where possible." The conformal interval is a finite-sample, distribution-free uncertainty band.

2. **EU AI Act (Article 13):** High-risk AI systems must enable users to "interpret the system's output and use it appropriately." A PD interval communicates the model's confidence directly.

Unlike bootstrap confidence intervals, the conformal guarantee does not rely on i.i.d. assumptions or asymptotic normality.

5.5 Practical Considerations

1. **Calibration set size.** With n calibration points, the coverage guarantee has a slack of $1/(n+1)$. For practical PD intervals, $n \geq 500$ per Mondrian group is recommended.

2. **Score function choice.** The coverage guarantee holds for *any* score function, but the interval *width* depends on the score. Scores that better reflect local uncertainty produce tighter intervals.

3. **Model quality matters.** Conformal prediction does not fix a bad model — it wraps it in wider intervals. A well-calibrated XGBoost will produce tighter conformal intervals than a poorly calibrated logistic regression. The conformal framework *rewards* model accuracy with narrower intervals while maintaining the coverage guarantee regardless.

4. **Through-the-cycle vs. point-in-time.** If the calibration set is from a benign period and the model is applied during a recession, exchangeability breaks. Section 6 addresses this with Adaptive Conformal Inference.

6 Adaptive Conformal Inference (ACI)

Financial data are non-stationary. Volatility clusters. Credit cycles span years. Regime shifts (the 2008 crisis, COVID-19, rate hikes in 2022) change the data-generating process abruptly. Under distribution shift, the exchangeability assumption breaks, and the static conformal guarantee no longer holds.

6.1 The Problem: Non-Stationarity

Consider applying a conformal credit model calibrated on 2019 data to applications arriving in March 2020. The calibration distribution (pre-COVID economy) differs from the test distribution (pandemic economy). The conformal scores from 2019 underestimate the model errors in 2020, and the prediction intervals systematically under-cover.

Warning

Static conformal prediction assumes calibration and test data are drawn from the same distribution (or at least are exchangeable). In finance, this assumption is always eventually violated. A conformal model deployed without adaptive recalibration will experience coverage breakdowns precisely when they matter most — during market stress.

6.2 The ACI Algorithm

Gibbs and Candès (2022) propose a simple online update rule that restores long-run coverage guarantees under arbitrary distribution shift.

Algorithm 4: Adaptive Conformal Inference (ACI)

Input: Initial miscoverage level $\alpha_1 = \alpha$, step size $\gamma > 0$, model $\hat{f}$, initial conformal scores $\{s_1, \ldots, s_n\}$.
For $t = 1, 2, 3, \ldots$**:**

1. Compute the adaptive threshold:

$$\hat{q}_t = \text{Quantile}\big(\{s_1, \ldots, s_n\};\ 1 - \alpha_t\big).$$

2. Produce the prediction interval: $\mathcal{C}_t(X_t) = [\hat{f}(X_t) - \hat{q}_t,\ \hat{f}(X_t) + \hat{q}_t]$.

3. Observe Y_t and compute the coverage error:

$$\text{err}_t = \mathbf{1}\{Y_t \notin \mathcal{C}_t(X_t)\}.$$

4. Update the miscoverage level:

$$\alpha_{t+1} = \alpha_t + \gamma \cdot (\alpha - \text{err}_t). \tag{13}$$

6.3 Intuition Behind the Update Rule

The update rule in (13) is a form of integral control:

- If $Y_t \notin \mathcal{C}_t(X_t)$ (a miss, $\text{err}_t = 1$): then $\alpha_{t+1} = \alpha_t + \gamma(\alpha - 1) < \alpha_t$. The effective significance level *decreases*, making the next interval *wider*.
- If $Y_t \in \mathcal{C}_t(X_t)$ (a hit, $\text{err}_t = 0$): then $\alpha_{t+1} = \alpha_t + \gamma\alpha > \alpha_t$. The effective significance level *increases*, making the next interval *narrower*.

Core Idea

ACI is a thermostat for coverage. When the model under-covers (too many misses), it widens intervals. When the model over-covers (too few misses), it tightens them. The step size γ controls the speed of adaptation: larger γ reacts faster to distribution shifts but introduces more variability in interval width.

6.4 Theoretical Guarantee

Theorem 6.1 (ACI long-run coverage (Gibbs & Candès, 2022))**.** *For **any** sequence of data $(X_1, Y_1), (X_2, Y_2), \ldots$ (no distributional assumptions whatsoever), the ACI procedure with step size $\gamma > 0$ satisfies:*

$$\left| \frac{1}{T} \sum_{t=1}^{T} \text{err}_t - \alpha \right| \leq \frac{|\alpha_1 - \alpha_{T+1}|}{\gamma T} + \frac{1}{T}. \tag{14}$$

In particular, if α_t remains bounded, the time-averaged miscoverage rate converges to α as $T \to \infty$.

Proof. Telescope the update rule. By (13):

$$\alpha_{T+1} - \alpha_1 = \gamma \sum_{t=1}^{T} (\alpha - \text{err}_t) = \gamma T \alpha - \gamma \sum_{t=1}^{T} \text{err}_t.$$

Rearranging:

$$\frac{1}{T}\sum_{t=1}^{T}\mathrm{err}_t = \alpha - \frac{\alpha_{T+1}-\alpha_1}{\gamma T}.$$

Since α_t is bounded (it can be projected onto $[0,1]$), the correction term $(\alpha_{T+1}-\alpha_1)/(\gamma T) \to 0$ as $T \to \infty$. □

Remark 6.1. The guarantee in Theorem 6.1 is **adversarial**: it holds for any data sequence, including one chosen by a malicious adversary who knows the algorithm. This is far stronger than the exchangeability requirement of static conformal prediction. The price is that the guarantee is *asymptotic* (long-run average) rather than *finite-sample* (each individual prediction).

6.5 Demonstration: Volatility Regime Shift

Example 6.1 (ACI during a vol spike)**.** Consider daily VaR prediction over 500 trading days. Days 1–250: low-volatility regime (annualized vol $\approx$ 12%). Days 251–300: a sudden volatility spike (annualized vol $\approx$ 40%). Days 301–500: gradual return to normal.

Method	**Coverage (days 251–300)**	**Recovery time**
Static conformal ($\alpha = 0.05$)	78% (severe under-coverage)	Never recovers
ACI ($\gamma = 0.01$)	86% (mild under-coverage)	~50 days
ACI ($\gamma = 0.05$)	91%	~20 days
ACI ($\gamma = 0.10$)	93%	~10 days

Static conformal, calibrated on the low-vol period, produces intervals that are far too narrow during the spike. ACI detects the under-coverage (many misses) and progressively widens intervals. With $\gamma = 0.05$, coverage recovers to the target level within approximately 20 observations.

6.6 Practical Choices

1. **Step size γ.** Typical values range from 0.005 to 0.05. A useful heuristic: set $\gamma = \alpha/10$ for a balance between reactivity and stability. For $\alpha = 0.05$, this gives $\gamma = 0.005$.

2. **Bounding α_t.** In practice, project α_t onto $[\epsilon, 1-\epsilon]$ (e.g., $\epsilon = 0.001$) to avoid degenerate intervals (empty or infinite).

3. **Rolling calibration window.** Instead of using a fixed calibration set from the past, maintain a rolling window of the most recent n observations. This combines ACI's adaptive threshold with fresh calibration data.

4. **Combining ACI with CQR.** Apply the ACI update to the CQR procedure: use quantile regression for interval shape and ACI for threshold adaptation. This provides both local adaptivity (CQR) and temporal adaptivity (ACI).

7 References

[1] **Angelopoulos, A. N., Bates, S., Candès, E. J., Jordan, M. I., and Lei, L.** (2024). Conformal Risk Control. *Journal of Machine Learning Research*, 25(260):1–42.

[2] **Vovk, V., Gammerman, A., and Shafer, G.** (2005). *Algorithmic Learning in a Random World.* Springer.

[3] **Romano, Y., Patterson, E., and Candès, E. J.** (2019). Conformalized Quantile Regression. *Advances in Neural Information Processing Systems (NeurIPS)*, 32.

[4] **Gibbs, I. and Candès, E. J.** (2022). Adaptive Conformal Inference Under Distribution Shift. *Advances in Neural Information Processing Systems (NeurIPS)*, 35.

[5] **Luo, R. and Colombo, N.** (2025). Conformal Prediction for Financial Time Series. *Advances in Neural Information Processing Systems (NeurIPS)*, 38.

[6] **Koenker, R. and Bassett, G.** (1978). Regression Quantiles. *Econometrica*, 46(1):33–50.

[7] **Lei, J., G'Sell, M., Rinaldo, A., Tibshirani, R. J., and Wasserman, L.** (2018). Distribution-Free Predictive Inference for Regression. *Journal of the American Statistical Association*, 113(523):1094–1111.

[8] **Angelopoulos, A. N. and Bates, S.** (2023). Conformal Prediction: A Gentle Introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591.

[9] **Basel Committee on Banking Supervision** (2005). *An Explanatory Note on the Basel II IRB Risk Weight Functions*. Bank for International Settlements.

[10] **Board of Governors of the Federal Reserve System** (2011). *SR 11-7: Guidance on Model Risk Management*. Office of the Comptroller of the Currency.